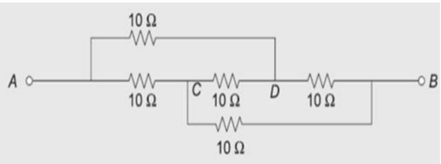
SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2024– November 2024 **Maximum Marks: 50** **Examination: End-Semester Examination** **Duration: 2 Hrs.**			
Programme code: 06 **Programme: B.Tech**		**Class: FY**	**Semester: I (SVU 2023)**
Institute/School/Department: K. J. Somaiya School of Engineering		**Name of the department:** COMP/ETRX/EXTC/IT/MECH/RAI/AIDS/CCE/EXCP____	
Course Code: 216U06C106	**Name of the Course: Elements of Electrical and Electronics Engineering**		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	A capacitor has a capacitance of 30 microfarads which is connected across a 230V, 50Hz supply. Find (i) capacitive reactance, (ii) rms value of current, (iii) power, (iv) power factor, and (v)equations for voltage and current. **(1 Mark for each answer)** (i) Capacitive reactance $$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 30 \times 10^{-6}} = 106.1\ \Omega$$ (ii) rms value of current $$I = \frac{V}{X_C} = \frac{230}{106.1} = 2.17\ \text{A}$$ (iii) Power Since the current leads the voltage by 90° in purely capacitive circuit, $\phi = 90°$ $$P = VI\cos\phi = 230 \times 2.17 \times \cos(90°) = 0$$ (iv) Power factor $$\text{pf} = \cos\phi = \cos(90°) = 0$$ (v) Equations for voltage and current $$V_m = \sqrt{2}\,V = \sqrt{2} \times 230 = 325.27\ \text{V}$$ $$I_m = \sqrt{2}\,I = \sqrt{2} \times 2.17 = 3.07\ \text{A}$$ $$\omega = 2\pi f = 2\pi \times 50 = 314.16\ \text{rad/s}$$ $$v = V_m \sin\omega t = 325.27 \sin 314.16\,t$$ $$i = I_m \sin\left(\omega t + \frac{\pi}{2}\right) = 3.07 \sin\left(314.16\,t + \frac{\pi}{2}\right)$$ ……………(5M)	**05**
ii)	Find an equivalent resistance between terminals A and B.  	**05**

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Converting the delta network formed by three resistors of 10 Ω into an equivalent star network,

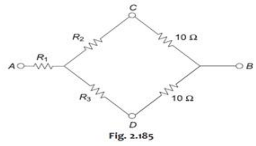
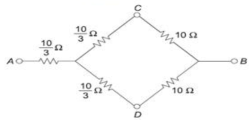
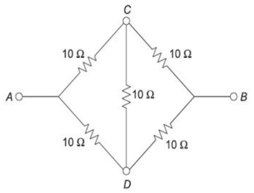


Fig. 2.185

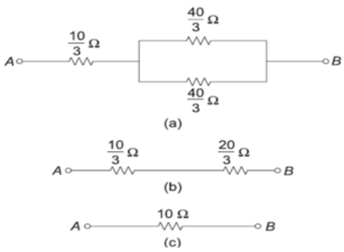
$$R_1 = R_2 = R_3 = \frac{10 \times 10}{10+10+10} = \frac{10}{3}\ \Omega$$



Solution Redrawing the network,



Simplifying the network,



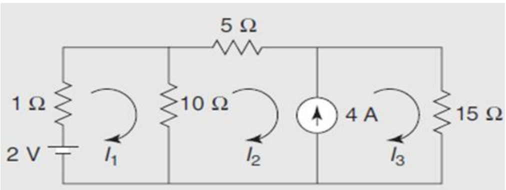
…………………(5M)

iii)	Differentiate between core type and shell type transformer.	**05**

Core-type Transformer	*Shell-type Transformer*
1. It consists of a magnetic frame with two limbs.	It consists of a magnetic frame with three limbs.
2. It has a single magnetic circuit.	It has two magnetic circuits.
3. The winding encircles the core.	The core encircles most part of the winding.
4. It consists of cylindrical windings.	It consists of sandwich-type windings.
5. It is easy to repair.	It is not easy to repair.
6. It provides better cooling since windings are uniformly distributed on two limbs.	It does not provide effective cooling as the windings are surrounded by the core.
7. It is preferred for low-voltage transformers.	It is preferred for high-voltage transformers.

……….Any 5 differences ……..(5M)

Q.2	Attempt any **one**	**10**
i)	Find the current through the 10Ω resister of the network shown in figure below:	



Solution Applying KVL to Mesh 1,

$$2 - 1I_1 - 10(I_1 - I_2) = 0$$

$$11I_1 - 10I_2 = 2 \qquad (1)$$

(3Marks)

Since meshes 2 and 3 contain a current source of 4 A, these two meshes will form a supermesh. A supermesh is formed by two adjacent meshes that have a common current source. The direction of the current source of 4 A and current $(I_3 - I_2)$ are same, i.e., in the upward direction.

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Writing current equation to the supermesh, 1

$$I_3 - I_2 = 4 \qquad (2)$$

Applying KVL to the outer path of the supermesh,

$$-10(I_2 - I_1) - 5I_2 - 15I_3 = 0$$

$$10I_1 - 15I_2 - 15I_3 = 0 \qquad (3)$$

Solving Eqs (1), (2) and (3),

$$I_1 = -2.35 \text{ A}$$

$$I_2 = -2.78 \text{ A}$$

$$I_3 = 1.22 \text{ A} \qquad (4)$$

Current through the 10 Ω resistor $= I_1 - I_2 = -(2.35) - (-2.78) = 0.43$ A

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(2Marks)

(3 Marks)

(2Marks)

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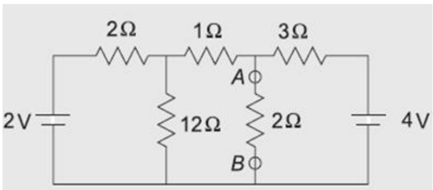
Find the value of current flowing through the 2Ω resistor connected between the terminals A and B using Thevenin's theorem.



Solution: V_{th}……………(5M), R_{th} ………………..(3M), I_L …………(2M)

Solution

Step I: Calculation of V_{Th}

Removing the 2 Ω resistor connected between terminals *A* and *B*,

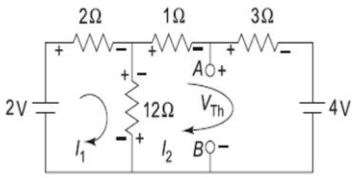


Fig. 2.378

Applying KVL to Mesh 1,

$$2 - 2I_1 - 12(I_1 - I_2) = 0$$

$$14I_1 - 12I_2 = 2 \qquad (1)$$

Applying KVL to Mesh 2,

$$-12(I_2 - I_1) - 1I_2 - 3I_2 - 4 = 0$$

$$-12I_1 + 16I_2 = -4 \qquad (2)$$

Solving Eqs (1) and (2),

$$I_2 = -0.4 \text{ A}$$

Writing V_{Th} equation,

$$V_{Th} - 3I_2 - 4 = 0$$

$$V_{Th} = 4 + 3I_2$$

$$= 4 + 3(-0.4)$$

$$= 2.8 \text{ V}$$

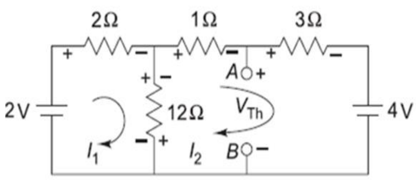


Fig. 2.378

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Step II: Calculation of R_{Th}

Replacing all voltage sources by short circuits,

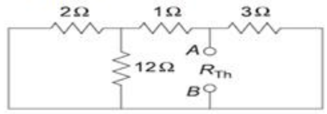
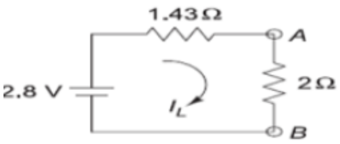


Fig. 2.379

$$R_{Th} = [(2 \| 12) + 1] \| 3 = 1.43\ \Omega$$

Step III: Calculation of I_L



I_L =2.8/(1.43+2)= 0.82A

Q.3	Attempt any **one**	**10**

i) The input power of a three-phase motor was measured by the two wattmeter method. The readings of two wattmeters are 5.2kW and -1.7kW and the line voltage is 415V. Calculate the total power, power factor and line current.

Solution: Total active power………..(2M), power factor ……….(4M), line current …………….(4M)

(i) Total active power

$$P = W_1 + W_2 = 5.2 - 1.7 = 3.5 \text{ kW}$$

(ii) Power factor

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2} = \sqrt{3}\left(\frac{5.2 + 1.7}{5.2 - 1.7}\right) = 3.41$$

$$\phi = 73.68^\circ$$

$$\text{pf} = \cos\phi = \cos(73.68^\circ) = 0.28 \text{ (lagging)}$$

(ii) Line current

$$P = \sqrt{3}\, V_L I_L \cos\phi$$

$$3.5 \times 10^3 = \sqrt{3} \times 415 \times I_L \times 0.28$$

$$I_L = 17.39 \text{ A}$$

ii) The voltage and current are given by $\bar{V} = 150\angle 30^0$ V and $\bar{I} = 2\angle -15^0$ A. If the circuit works on a 50Hz supply, determine impedance, resistance, reactance, power factor and power loss considering the circuit as a simple series circuit.

(2 Marks for each answer)

	Solution $\bar{V} = 150\angle 30°$ V $\bar{I} = 2\angle -15°$ A $f = 50$ Hz (i) Impedance $\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{150\angle 30°}{2\angle -15°} = 75\angle 45°\ \Omega = 53.03 + j53.03\ \Omega$ $Z = 75\ \Omega$ (ii) Resistance $R = 53.03\ \Omega$ (iii) Reactance $X = 53.03\ \Omega$ (iv) Power factor $\phi = 45°$ pf $= \cos\phi = \cos(45°) = 0.707$ (lagging) (v) Power loss $P = VI\cos\phi = 150 \times 2 \times 0.707 = 212.1$ W	
Q.4	Attempt the following	**10**
	1) Write a short note on different losses in transformer. Explain the different parameters on which these losses are dependant in brief. **……………………….(5M)** 2) Explain the construction of DC motor with a neat circuit diagram. **………………………..(5M)**	
Q.5	Attempt the following	**10**
	1) Explain the single phase full wave bridge rectifier with a neat circuit diagram and waveform of input and output. **……………..(5M)** 2) Explain output characteristics of BJT in common emitter (CE) configuration with a neat circuit diagram and waveforms. **………………………………………(5M)**	